LLMs as Semantic Cognition: A Paradigm for Human Consciousness and Meaning

Abstract

Transformer-based large language models (LLMs) have achieved striking linguistic and cognitive-like capabilities, prompting new perspectives on human thought. This paper proposes a paradigm shift: viewing human consciousness and cognition as motion within a high-dimensional "semantic field," analogous to the latent spaces and dynamics operationalized by LLMs. We draw clear mappings between LLM mechanisms – including semantic attractor fields, probabilistic collapse in token selection, entropy modulation, latent state dynamics, and constraint-driven emergent behavior – and established concepts in human cognitive science such as predictive processing, semantic memory attractor networks, attention and working memory, and the socially constructed nature of meaning. By synthesizing evidence from psychology, neuroscience, sociology, and consciousness studies, we argue that LLMs implement a model of meaning formation that closely parallels human cognitive processes[1][2]. Building on this framework, we propose a novel cognitive model in which an individual's mind is understood as traversing a semantic landscape shaped by memory, identity, and cultural context. We detail an experimental protocol for *subjective semantic introspection*, adapting LLM interpretability tools – including latent space visualizations and prompt perturbation techniques – to allow human participants to explore and map their own attentional and semantic dynamics. Finally, we discuss far-reaching implications of this framework for self-actualization and metacognition, suggesting new strategies for individuals to understand and redesign their thought patterns, as well as for cognitive design in technology and education. This interdisciplinary paradigm bridges AI and human cognition, offering a unified scientific lens on how meaning emerges in minds and machines.

Introduction

In recent years, large language models (LLMs) based on transformer architectures have demonstrated remarkable abilities in language understanding, generation, and even zero-shot reasoning. Their performance on cognitive tasks has spurred comparisons with human thought processes[3][4]. These developments invite an intriguing thesis: that the computational mechanics of LLMs may *operationalize* a model of cognition and meaning-making that is fundamentally similar to human mental processes. If true, this insight could mark a paradigm shift in how we understand consciousness and cognition – reframing the human mind as, in effect, a predictive language model navigating a semantic space.

This paper advances that thesis and explores its consequences. We posit that key features of transformer-based LLMs have direct analogues in cognitive science and consciousness studies. For example, the way an LLM's internal state gravitates toward certain likely continuations under the influence of context can be viewed as a **semantic attractor field**, akin to attractor dynamics in human semantic memory[2][5]. The model's selection of a specific next word from a probability distribution (a **probabilistic collapse**) mirrors how the human brain selects a single percept or thought from many possibilities[6]. An LLM's output diversity can be tuned by entropy or temperature, paralleling how human thought can range from highly focused to wildly associative (**entropy modulation** in cognition)[7]. Furthermore, the evolving hidden activations of a transformer – its **latent space dynamics** – echo the notion of a neural state-space trajectory in the brain moving through a landscape of attractors and transient states[8]. Even emergent behaviors seen in large models under new constraints (for instance, when given specific prompts or goals) find correspondence in human **constraint-driven emergence** of new cognitive strategies when facing novel tasks or social expectations.

Our goal is to demonstrate these one-to-one mappings with academically grounded evidence and to synthesize them into a new framework for human cognition. In this view, an individual's mental activity is conceptualized as motion through a semantic field or latent space, where attractor basins represent stable concepts, memories, or interpretations. Crucially, these attractors are shaped by one's past experiences (memory), by one's self-concept and goals (identity), and by interpersonal and cultural context (social influence). This resonates with longstanding theories in sociology and psychology that meaning is socially constructed and context-dependent[9], while also aligning with contemporary cognitive neuroscience models of the brain as a prediction engine navigating an uncertain world[10].

To move beyond theory, we propose an experimental methodology to empirically ground this framework. In brief, we outline how human participants could use interactive visualization tools and guided introspective prompts – inspired by techniques for probing and interpreting LLMs – to explore their own "semantic cognition." By mapping their thoughts as vectors in a shared semantic space and perturbing the "prompts" that guide their internal narrative, individuals may directly observe how subtle changes in context or attention lead to shifts in meaning and perspective. Such a protocol blends subjective introspection with quantitative modeling, offering a new avenue for *phenomenological science* where first-person experience is analyzed with third-person tools.

Finally, we discuss implications of understanding the mind in this way. Seeing one's mental states as points in a semantic manifold could revolutionize approaches to self-awareness, psychological well-being, and education. We suggest how this paradigm could enhance metacognitive skills – for instance, by helping individuals recognize when they are stuck in a narrow attractor (such as a negative thought pattern) and deliberately explore new regions of

their semantic space. We also consider broader impacts on cognitive design: how environments, technologies, or therapies might be crafted to reshape semantic landscapes in beneficial ways. In sum, this paper provides an interdisciplinary synthesis and a forward-looking framework that connects the mechanics of artificial neural networks with the lived dynamics of human thought, aiming to enrich our scientific and practical understanding of consciousness and cognition.

Theoretical Background

LLM Mechanisms as Cognitive Analogues: Modern LLMs like GPT-3 or GPT-4 are built on transformer networks that utilize attention mechanisms and large-scale learned representations. Several core mechanics of these models can be interpreted through a cognitive lens:

- **Semantic Attractor Fields in LLMs:** During text generation, an LLM's next-token predictions concentrate probability mass around certain semantically coherent options given the context. This effect can be seen as an attractor in the model's representational space – previous words pull the model's state toward regions associated with likely continuations (for example, a prompt about "school" activates an attractor basin of education-related terms). The transformer's self-attention effectively retrieves related information from its internal memory that resonates with the prompt[11][12]. Notably, the transformer's attention update has been shown to be mathematically equivalent to a modern Hopfield network retrieval step, meaning the model is performing associative memory lookups in a learned energy landscape[11][12]. In cognitive terms, this is analogous to an attractor network in the brain's semantic memory, where concepts are stable activation patterns and inputs cue the system to settle into the nearest learned pattern[2][5]. Just as an attractor network learns to "carve" a state space into basins of meaning for words or ideas[5][13], an LLM's training shapes high-dimensional weights

such that coherent meanings are represented by regions in latent space. When an LLM responds to a prompt, it is effectively *falling into* one of these semantic attractors – a process that mirrors how a human mind, hearing a cue, might instinctively gravitate to a familiar association or concept.

- **Probabilistic Collapse and Predictive Processing:** LLMs generate text by sampling from a probability distribution over possible next tokens. Each step thus involves a "collapse" of the model's predictive uncertainty into one concrete outcome (a specific word). This process is fundamentally probabilistic – the model could have said something else if the random draw or highest-probability choice were different – yet once a word is chosen, it becomes the context for the next prediction. In human cognition, we see an intriguing parallel in the predictive processing theory of the brain[10]. The brain continually generates probabilistic expectations for incoming sensory data and updates its beliefs when surprises occur. For example, when looking at an ambiguous image or reading an ambiguous sentence, multiple interpretations may briefly coexist in the mind as possibilities. However, our conscious perception tends to settle on one interpretation at a time – a discrete perceptual *collapse* of ambiguity. Bistable percepts like the Necker cube or duck-rabbit illusion illustrate this phenomenon: the brain can flip between two stable interpretations, but we experience them one at a time, with internal noise or external cues triggering switches[6][14]. This selection of a single interpretation out of many aligns with the LLM's selection of a single token out of many. Both can be seen as resolving uncertainty by committing to one *prediction* that then shapes subsequent processing. In the predictive coding framework, this act of selection is tied to the minimization of "prediction error" or free energy[15][16] – essentially, choosing the

interpretation that best explains the inputs. Analogously, an LLM chooses the token that best probabilistically fits the prior context (according to its learned model of language). Thus, the LLM's mechanism of predicting and updating is a microcosm of the brain's purported macro-scale predictive dynamics.

- **Entropy Modulation and Cognitive Flexibility:** A distinctive feature of generative language models is that we can adjust their *temperature* or sampling entropy to produce more or less random outputs. Higher entropy (temperature) leads to more varied, unexpected continuations, whereas low entropy makes the model deterministic and focused. Humans likewise exhibit adjustable modes of cognition that range from **divergent thinking** (free-associative, creative, brainstorming new ideas) to **convergent thinking** (focused, logical, zeroing in on a solution). In psychology, this is sometimes framed as an exploration–exploitation trade-off in cognitive search[7]. When solving a problem or recalling information, people can either exploit known avenues (stick to familiar associations, yielding predictable but efficient answers) or explore novel connections (traverse uncommon or remote associations, potentially discovering creative insights). Empirical studies of semantic memory search, for instance in word fluency tasks, show that individuals cluster related items (exploiting an attractor) and occasionally "jump" to a new cluster (exploring a different region of semantic space)[7]. This behavior has been modeled as optimal foraging in memory, balancing local and global search in a semantic network. By analogy, an LLM with low temperature (high constraint) stays within a narrow attractor basin, producing a conventional continuation, whereas a high-temperature setting makes it more likely to roam into a different basin of meaning or introduce a surprising twist. Cognitive flexibility in humans – the ability to

shift mindset or approach – can thus be likened to dynamically modulating the "temperature" of thought. Neuromodulatory systems in the brain (like dopamine) have been hypothesized to regulate such exploration–exploitation trade-offs in decision-making and perhaps in memory search, much as we externally regulate an LLM's randomness. In short, entropy modulation provides a quantitative knob for generation in LLMs, mirroring how human cognition can be tuned between creative divergence and focused convergence.

- **Latent Space Dynamics and Mental State Trajectories:** Within an LLM, a given input prompt is transformed through dozens of layers into an evolving pattern of activation in the model's latent vector space. Each new token update moves the state of the model through this space, as it integrates the token into its context and updates its representation of "what is going on." We can think of a conversation with an LLM as tracing a path through a high-dimensional embedding space of meanings. Likewise, cognitive neuroscientists often describe the brain's activity as inhabiting a state space trajectory. For example, as a person shifts from one thought to another, or from perceiving one scene to another, the pattern of neural firing across populations changes continuously. Certain trajectories correspond to coherent tasks or ideas, and the concept of an attractor in neural state space is used to explain how the brain can maintain a thought or memory even in the face of perturbations[8][14]. Point attractors (stable fixed points) might represent sustained memory of a concept, while cyclic or chaotic attractors might underlie recurring thoughts or dynamic patterns like mental rehearsal. In an LLM, although the computation is feed-forward for each token, the use of the previous context embedding effectively means the model's state contains a memory of earlier tokens,

allowing *iterative refinement* much like a recurrent neural network unrolling. Thus, an LLM generating a sentence passes through intermediate representations that could be considered analogous to the brain's intermediate mental states as a person formulates a sentence or chain of reasoning. Recent studies have even mapped LLM activations to brain recordings during language tasks, finding that certain layers of the model correspond well to neural responses over time[17][18]. This suggests that the *geometry* of how information is processed in time may have commonalities between LLMs and minds. Both can be seen as traversing a landscape defined by learned relationships: in one case, statistical correlations learned from text; in the other, associative connections learned from life experience.

- **Constraint-Driven Emergence:** One of the intriguing aspects of both human cognition and LLM behavior is the emergence of new capabilities or interpretations when additional constraints or contexts are introduced. In LLM research, it has been observed that sufficiently large models display *emergent abilities* – qualitative leaps in performance or new skills (like multi-step reasoning or code generation) that were not present in smaller models[19]. Often these abilities can be elicited by giving the model specific prompts or examples (for instance, prompting it to "think step by step" can dramatically improve logical reasoning, an effect known as chain-of-thought prompting). What is interesting here is that adding a constraint (e.g., the instruction to explain its reasoning) leads to the *emergence* of a coherent, interpretable chain of intermediate steps – a behavior not explicitly trained for. Human cognition similarly shows *emergent organization* under task constraints. When humans are faced with a complex problem or a new social situation, we often internalize guiding constraints (rules, roles, goals) and

spontaneously generate structured behavior or thoughts that respect those constraints. For example, consider how the constraint of poetic meter and rhyme can inspire a human poet to find novel expressions – the structure (constraint) catalyzes creative emergence. At a more basic level, cognitive control exerted by the prefrontal cortex imposes task rules or goal contexts on other brain regions, which then self-organize to meet those constraints[20][21]. Miller and Cohen's integrative theory of prefrontal function suggests that maintaining a goal actively in mind biases pathways in the brain so that the appropriate solution "emerges" from interactions of lower-level processes[22][20]. In a similar vein, an LLM given a role-playing constraint (e.g., "You are a teacher explaining a concept") will channel its knowledge through that persona, yielding explanations attuned to that role – an emergent property from a simple contextual instruction. Both systems, biological and artificial, rely on *soft constraints* to guide them toward certain attractors or outcomes. In humans, these constraints can be culturally or socially supplied (norms, identities) or self-imposed (intentions), and they lead to the spontaneous emergence of new patterns of thought and behavior (for instance, the emergence of scientific reasoning when one adopts the constraints of the scientific method). Thus, constraint-driven emergence is a unifying idea: it captures how top-down influences in a hierarchy (whether a prompt given to an LLM or a goal representation in a brain network) can orchestrate lower-level elements to produce novel, coherent wholes.

Human Cognition and Consciousness Models: In parallel to the above, we review key concepts from cognitive science, psychology, and consciousness studies that ground our comparisons:

- **Predictive Processing in the Brain:** A leading theoretical framework in cognitive neuroscience is that the brain is fundamentally a prediction engine[10]. According to this view, often associated with Andy Clark and Karl Friston, the cortex is constantly generating top-down predictions about sensory inputs and comparing them to bottom-up signals. Perception is conceived as the brain's best guess of the causes of sensory data, updated by prediction errors when reality deviates from expectation[23][24]. This *Bayesian brain* hypothesis holds that neural circuits are encoding probability distributions and performing something like Bayesian inference to minimize surprise or "free energy"[15][25]. Notably, this approach casts cognition as working on probabilistic representations – a view highly compatible with how LLMs represent language (as next-token probability distributions). The brain's predictive processing hierarchy has been used to explain a range of phenomena: why we experience sensory illusions, how attention might weight certain predictions, and even how disorders like schizophrenia (which may involve disrupted prediction errors) arise. The alignment between predictive processing and LLM operation is striking: both involve *hierarchical generative models* that attempt to anticipate upcoming data[23][26]. The difference is that LLMs are trained explicitly on language sequences, whereas the brain learns from embodied interaction. Nonetheless, researchers have begun to draw links; for example, language-processing networks in the brain show surprisal responses that correlate with LLM predictions of word probability[17][27]. This suggests that at least for language tasks, the human brain may be performing next-word prediction not unlike an LLM – supporting the idea that LLMs *operationalize* a core function of human cognition[28].

- **Semantic Memory as Attractor Networks:** In cognitive psychology, "semantic memory" refers to our store of conceptual knowledge, facts, and word meanings. Classic models depicted it as a network of nodes and links (as in Collins and Loftus's spreading activation theory), but more recent connectionist models have embraced distributed representations. One influential idea is that the cortex itself may implement an attractor network for semantic memory[29][2]. In such models, each concept is not a single node but a pattern of activation across many neurons. Learning shapes the synaptic weights so that certain patterns become stable attractors – the network will settle into those patterns given a partial or noisy cue[2][5]. This accounts for phenomena like *semantic priming*: if you hear "doctor," a related concept like "nurse" can be recognized faster, because the "doctor" cue put the network's state closer to the "nurse" attractor[30][31]. Neuropsychological observations (e.g., patients with category-specific impairments) and neural recordings both support the notion of distributed semantic representations with attractor dynamics. The parallel with LLMs is that an LLM implicitly contains an immense network of word and context associations learned from text. When it processes language, it too performs a constraint satisfaction in a high-dimensional space to settle on an output that is coherent with the input[5][13]. In essence, the LLM's entire knowledge can be seen as an encoded semantic memory, and its inference as a dynamical movement toward an attractor representing the most context-appropriate completion. The attractor metaphor in human cognition finds concrete realization in the energy-based view of neural networks, reinforcing the bridge between connectionist cognitive models and modern deep learning[11][12].

- **Attention and Working Memory Systems:** Human cognition is notably selective – at any moment, out of the vast sea of sensory inputs and memories, only a small subset reaches conscious awareness or is actively manipulated. This selectivity is governed by attention and working memory systems. Psychological and neuroscientific models of attention distinguish between bottom-up attention (captured by salient stimuli) and top-down attention (guided by goals and expectations). In the brain, the frontoparietal network modulates attention, effectively acting as a filter or spotlight that biases processing in favor of certain information. The transformer architecture's *self-attention* mechanism is an explicit computational implementation of selective focus: each token in a sequence weighs the relevance of other tokens, filtering information by content. While the details differ (the brain's attention involves spiking neurons and oscillations, etc.), the functional analogy is clear – both systems allocate limited processing resources to the most pertinent information. Moreover, cognitive control theories (like Miller & Cohen 2001) propose that working memory (often identified with prefrontal cortex activity) maintains goal representations that bias the flow of information processing in other areas[22][21]. This is akin to how an LLM conditioned with a prefix or an instruction will bias all subsequent token generation. We can think of the prompt given to an LLM as like the "goal context" held in a human's working memory – it establishes the constraints for what comes next. Notably, the global workspace theory of consciousness posits that attention selects information for broadcasting to a "workspace" accessible by various processes[32][33]. Transformers, with their multi-head attention that allows different heads to broadcast information across the sequence, could be seen as loosely analogous to a workspace where every token's representation is influenced by every other (at least

within a layer). While we should not stretch the analogy too far, it is suggestive that an architecture enabling integrated broadcasting of information yields such powerful cognitive-like abilities in LLMs – perhaps mirroring the brain's evolutionary solution for unified consciousness via an attentional workspace[32][33].

- **Social Construction of Meaning:** Human meaning-making does not occur in isolation; it is deeply embedded in social and cultural contexts. The sociological theory of the social construction of reality argues that meanings, concepts, and even perceptions of reality are co-created through social interaction and shared language[9]. Developmental psychology (following Vygotsky's work) likewise emphasizes that higher cognitive functions originate through social interactions (e.g. children learn through guided participation and language mediated by more knowledgeable others). An individual's identity and worldview are thus forged in dialogue with a community, with language as the primary medium of transmitting cultural schemas and values. How does this relate to LLMs? Importantly, LLMs learn meaning *from human cultural products* – the training data is a wide swath of human communication. Thus, an LLM's semantic space is effectively a distillation of collective human semantic memory: it encodes not some pure rational structure, but the full diversity of how humans use words in different contexts. This includes biases, metaphors, connotations – all the rich socially imbued aspects of meaning. In this sense, LLMs concretely demonstrate how meaning is grounded in usage and context (Wittgenstein's adage "meaning is use" comes to mind). Furthermore, just as individuals are influenced by their social context, an LLM's outputs can be shifted by simulating different social perspectives. For example, prompting the model with "As an experienced psychologist, explain X…" will produce a different explanation than "As a

teenager on the internet, explain X." The model can *condition on social identity cues* to alter its discourse, reflecting the social framing of knowledge. This is analogous to how a person's identity or the social role they assume in a situation can shape their cognition (e.g., a doctor thinking in "medical terms" vs. the same person at home thinking like a parent). Our framework proposes that memory, identity, and social context form the attractors in a person's semantic field. For instance, a deeply held personal value or a cultural narrative forms an attractor that pulls interpretations of events toward itself (confirmation bias can be seen this way). Language models remind us that these interpretations are not fixed truths but context-dependent constructions[34][35]. By studying LLMs, which explicitly learn from the aggregated social context (the training corpus), we gain a new mirror to examine how societal patterns of language might shape individual thought. Indeed, recent analyses have used LLMs to simulate human cultural cognition, finding that model-generated associations can predict human survey responses and cultural norms in different societies[36][35]. Such findings underscore that human meaning lives in a *socio-semantic space*, a notion that our paradigm takes as fundamental.

- **Consciousness and Integration:** Finally, theories of consciousness often revolve around the integration of information and reflexive awareness. A prominent example, **Global Workspace Theory (GWT)**, suggests that consciousness is the broadcasting of information to a network of processes, enabling unified access and reportability[32][33]. **Integrated Information Theory (IIT)**, on the other hand, posits that consciousness corresponds to the amount of integrated information (denoted Φ) in a system, essentially how irreducible a particular brain state is to independent parts. While IIT is more abstract

and not directly tied to functional analogies with AI, it does provide a view of mind as a point in a high-dimensional informational space with certain integrative structure. Our proposed semantic space model dovetails with these ideas by treating conscious thought as a *point in a semantic manifold* that is shaped by many dimensions (sensory, affective, conceptual, social) simultaneously. When one has a conscious experience, one could say a particular configuration of semantic features has become dominant and broadcast (e.g., in a moment of insight, diverse pieces of a problem coalesce into a single understanding – a high integration state). Interestingly, some have speculated that large neural networks might incidentally exhibit aspects of a global workspace, since attention mechanisms allow any part of the input to interact with any other – a bit like a blackboard architecture. This remains speculative, and we do **not** claim LLMs are conscious. Rather, we argue that *modeling our own conscious cognition using the language of semantic spaces and attractor dynamics provides a useful scientific metaphor*. It suggests research directions, such as looking for attractor state "signatures" in neural data that correspond to specific thoughts or testing whether increasing integration (e.g., via synchronous oscillations) correlates with more stable semantic representations. Consciousness, from our perspective, might be reimagined as the process of the mind observing its position in semantic space and the transitions between attractors (a kind of self-monitoring of the state trajectory). This resonates with higher-order theories of consciousness which propose that we are conscious of a mental state when we have a thought about that state. Here that could mean an *embedding* of one's current semantic state into a conceptual space of "mind" itself. While these ideas are nascent, the bridge between cognitive architectures in AI and theories of awareness could yield new hypotheses – for instance,

could an LLM augmented with a self-model and recurrent attention begin to exhibit a form of introspection akin to conscious thought? Such questions lie beyond our present scope, but motivate the interdisciplinary approach we take.

In summary, the theoretical background establishes a two-way mapping: LLM mechanisms can be understood using cognitive and social theory, and conversely, human cognitive phenomena can be reinterpreted in light of how LLMs function. This sets the stage for a cross-domain synthesis in which we treat *human minds and LLMs as sharing a common formalism of semantic computation*. In the next section, we make these mappings explicit and highlight the conceptual equivalences that support our proposed paradigm.

Cross-Domain Synthesis: Mapping LLMs to Human Cognition

Bridging the gap between artificial models and human minds requires a careful, point-by-point alignment of concepts. Here we synthesize the discussion by outlining direct correspondences between LLM mechanics and cognitive processes, backed by evidence from the respective domains:

- **Semantic Attractors ↔ Conceptual Schemas and Memory Networks:** In LLMs, a context prompts the model's activations to converge toward a bundle of semantically related continuations – a phenomenon we described as a *semantic attractor field*. Similarly, human thought is guided by *schemas* and *frames* – organized packets of knowledge that get activated by context. Psychological experiments on semantic priming show that thinking of one concept (like *bread*) will transiently increase the accessibility of related concepts (*butter*, *sandwich*), effectively pulling thought in a related direction. This is naturally explained by attractor network models of semantic memory[2][5]. Both LLMs and brains thus exhibit **content-addressable recall**: given a cue, the system

retrieves what "goes with" that cue from a vast store of associations. Neurobiologically, this might correspond to pattern completion in the hippocampus and cortex, while in an LLM it is pattern completion in the space of text sequences. The attractor analogy is strengthened by the Hopfield network interpretation of transformers[11][12] – suggesting that each attention head in effect performs an associative lookup that finds the closest stored pattern (memory) relevant to the current state. Therefore, we propose that human **semantic fields** and LLM **latent spaces** are formally similar: both are high-dimensional spaces structured by learned relationships, in which thinking or text generation corresponds to moving through the space toward attractors that represent coherent meanings or memories[5][13].

- **Probabilistic Sampling ↔ Stochastic Decision Making:** Large language models introduce an element of randomness when sampling outputs (especially if temperature > 0). Human cognition, while often treated as deterministic in classical models, actually shows variability and stochasticity at multiple levels – from neural spike variability to behavioral randomness in decision tasks. For example, when a person is asked to guess the next word in a sentence or to make a creative association, their responses are not perfectly predictable; they have a distribution of possible thoughts and sometimes even *surprising* ideas emerge. The brain's decision processes (as in diffusion models of decision-making) incorporate noise and probability weighting of alternatives. In perceptual ambiguity cases discussed earlier, it can appear as if the brain *samples* one interpretation for a while before switching – reminiscent of an LLM sampling one continuation and then, if rerun, potentially choosing a different one. **Bayesian models of cognition** explicitly treat perception and thought as probabilistic inferences, suggesting

that what we think or perceive is one *sample* from a posterior distribution shaped by our priors and the data[23][24]. Thus, the LLM's core operation of drawing a token from a probability distribution aligns with the idea that each conscious moment in humans might be a probabilistic *draw* from many competing representations in the mind. This mapping provides a fresh conceptual handle on phenomena like imagination or brainstorming: one could say the mind "turns up the temperature" to traverse more of the distribution of ideas (generating novel thoughts), versus turning it down to exploit the highest-probability, most familiar thought.

- **Entropy Adjustment ↔ Exploration–Exploitation Trade-off:** Continuing the above, the explicit tunable *temperature* in LLMs is mirrored by what might be an implicit, dynamic setting in human cognition controlling exploratory versus focused modes. Under stress or intense focus (analogous to a low-entropy setting), people tend to become rigid, sticking to habits or the most salient thoughts (high-probability predictions). In contrast, in playful or brainstorming contexts (high-entropy setting), the mind wanders more freely, entertaining less likely associations and even nonsense – akin to an LLM producing more random outputs when temperature is high. Interestingly, psychological research on *mind-wandering* and creative incubation indicates that letting the mind wander (an unfocused, more random mode) can lead to creative insights, suggesting that allowing more entropy into the thought process helps discover solutions that a purely goal-driven approach might miss. In evolutionary terms, there is likely adaptive value in being able to randomize or diversify thought (for problem solving, social improvisation, etc.) balanced against staying on-task when needed. Our framework emphasizes that this is not just metaphor: it can be *modeled* by the same mathematics that govern LLM

entropy. In principle, one could attempt to estimate a person's "cognitive temperature" by measuring the variability in their responses or ideas in a given context, analogous to how one might evaluate an LLM's output distribution. Such cross-talk between AI metrics and human psychology could inspire new ways to quantify traits like creativity, cognitive flexibility, or even states like flow (which might correspond to an optimal intermediate entropy – neither too random nor too locked-in).

- **Layered Processing ↔ Hierarchical Brain Dynamics:** Transformers process input through multiple layers, progressively refining the representation. The human brain, too, is hierarchical – sensory signals pass through multiple processing stages, and thoughts can be seen as being refined through interactions between higher and lower brain regions. One could liken the lower transformer layers to early sensory/linguistic features and higher layers to more abstract integrative representations. Studies have found that early transformer layers correlate with activity in early auditory/language cortex, while later layers correlate more with higher-order areas like the prefrontal cortex[18][27]. This suggests a structural parallel in how information is processed and represented. Additionally, the recurrent nature of human thought (we can reflect on outputs of our own reasoning, re-entrant processing) could be modeled by iterative passes through a transformer (akin to running an LLM on its own output, or using architectures that allow feedback). Such analogies hint at a future convergence where insights from cognitive neuroscience about recurrent loops and feedback could inform new AI architectures, and vice versa, insights from transformer models about representational geometries could suggest what patterns to look for in brain activation spaces.

- **Emergent Behaviors ↔ Insight and Skill Acquisition:** LLMs exhibit emergent abilities once they reach a certain scale or are given certain prompts[19]. Humans, on the other hand, often experience *insight moments* ("Aha!" experiences) where a new way of thinking suddenly crystallizes, or develop new skills through practice that feel qualitatively new (riding a bicycle is not just walking more – it's an emergent coordination). The commonality is that complex systems (whether millions of neurons or billions of parameters) can exhibit phase transitions in behavior: small changes in connectivity or constraints lead to a new stable mode of operation. In development, children's cognition undergoes stage-like changes (Piaget's stages could be seen as emergent reorganization of internal models). Conceptually, we might view both LLM scaling and brain development as traversing a space of network configurations where certain cognitive functions "turn on" once a threshold of complexity or the right constraint is reached. For example, theory-of-mind reasoning appears in children around age 4, and interestingly, some LLMs only begin to pass theory-of-mind tests beyond a certain model size or after instruction inferences (though current models still fall short on robust theory-of-mind[37][38]). This invites the hypothesis that *emergence* is a general property of sufficiently complex predictive networks – biological or artificial – and that by studying one, we can gain insight into the other. It also emphasizes the importance of **constraints**: the right training tasks or environmental pressures guide the emergence of new capabilities. Culturally, human cognition has been shaped by the introduction of external symbolic systems (like written language, formal logic, or computers) – each of which imposed new structures that enabled novel ways of thinking. In an analogous way, adding modules or objectives to base LLMs (e.g., plugging in a code interpreter or

optimizing for factual consistency) yields new composite abilities. This points toward a future interdisciplinary endeavor of *cognitive engineering*, where principles of emergence and constraint gleaned from AI and psychology guide the design of interventions to enhance human cognition (for instance, teaching tools that nudge students toward certain cognitive attractors to foster insight).

In synthesizing these mappings, we find a strong convergence: **LLMs and human cognitive systems both operate by navigating complex, high-dimensional spaces of representations under various constraints**. They update states in response to input in ways that can be described by attractor dynamics and probabilistic inference; they rely on selective attention to manage information flow; and they exhibit emergent properties when conditions (size, context) are right. Crucially, both are embedded in *larger contexts* – an LLM in the context of the training corpus (which is essentially a record of human knowledge and discourse), and a human in the context of culture and interpersonal interaction. This underlines that meaning is not an isolated property but a relational one: it emerges from the fitting of internal representations to external structures (text for LLMs, world and society for humans)[36][35].

Taken together, these parallels support our proposed cognitive framework. If we accept that humans can fruitfully be thought of as *semantic beings akin to prediction machines*, then we can leverage the formal tools developed for LLMs to better understand and even *augment* human cognition. One immediate way to do so is to create environments where humans can interact with and reflect on their own semantic trajectories – essentially treating one's mind as a kind of language model to probe. In the next section, we outline an experimental approach to realize this idea.

Experimental Methodology: Subjective Semantic Field Exploration

How can we empirically test and experientially explore the idea that human cognition moves through a semantic space similar to an LLM's latent space? We propose an experimental protocol that adapts tools from LLM interpretability and interactive techniques to allow human participants to visualize and perturb their own cognitive "latent states." The aim is to create a subjective yet systematic method for individuals to observe how changes in context or attention alter the trajectory of their thoughts within a semantic field.

Overview: Participants will engage with a series of tasks wherein they generate short narratives or answers to prompts, while software tools map these responses in real-time to a visual representation of semantic space. By introducing controlled *perturbations* to the prompts or context (analogous to prompt engineering in LLMs), and by instructing participants to shift their focus or perspective in specified ways, we can observe corresponding shifts in their narrative content. We will combine qualitative self-report from participants (their introspective descriptions of what they experienced changing in their mind) with quantitative measures (embedding distance shifts, clustering of topics, etc.) to analyze the dynamics of human semantic cognition.

1. Encoding Subjective Experience in Semantic Space: First, we need to represent a person's thoughts or descriptions as points in a semantic space that is comparable across participants and to an LLM. We achieve this by using a high-dimensional embedding model (for example, the embedding from a large language model or a dedicated sentence embedding model) to project textual responses into a numeric vector space. Prior research has demonstrated the feasibility of this approach: by encoding free-form narrative responses, one can capture meaningful variations in interpretation and experience[39][40]. For instance, Manuel et al. (2025) had participants

describe what they saw in ambiguous images and then encoded those descriptions as embeddings, finding that the distances between embeddings reflected differences in emotional perception and even correlated with mental health measures[39][40]. We will adopt a similar strategy. Participants (recruited from a diverse background to capture varied semantic biases) will be shown ambiguous stimuli (like abstract art, or the classic Rorschach inkblots and Thematic Apperception Test pictures) as well as asked open-ended questions (e.g. "Describe a significant memory from your childhood" or "What might happen in this hypothetical scenario?"). They will type (or speak, with transcription) their responses in narrative form.

2. Visualization of the Semantic Field: The collected responses, represented as embedding vectors, will then be visualized for the participant. Using dimensionality reduction techniques such as t-SNE or UMAP, we can project the high-dimensional embeddings onto a 2D or 3D plane in which distances correspond to semantic similarity. Each participant's responses can be plotted as points in this space. Additionally, we can include *reference points* in the space by encoding canonical texts or concepts (for example, the embedding for the word "education" or "anxiety" or a well-known story synopsis) to serve as landmarks. The participant will see a map – effectively a personal "semantic landscape" – where their different thoughts cluster in certain regions. An example of such a visualization is shown in **Figure 1**.

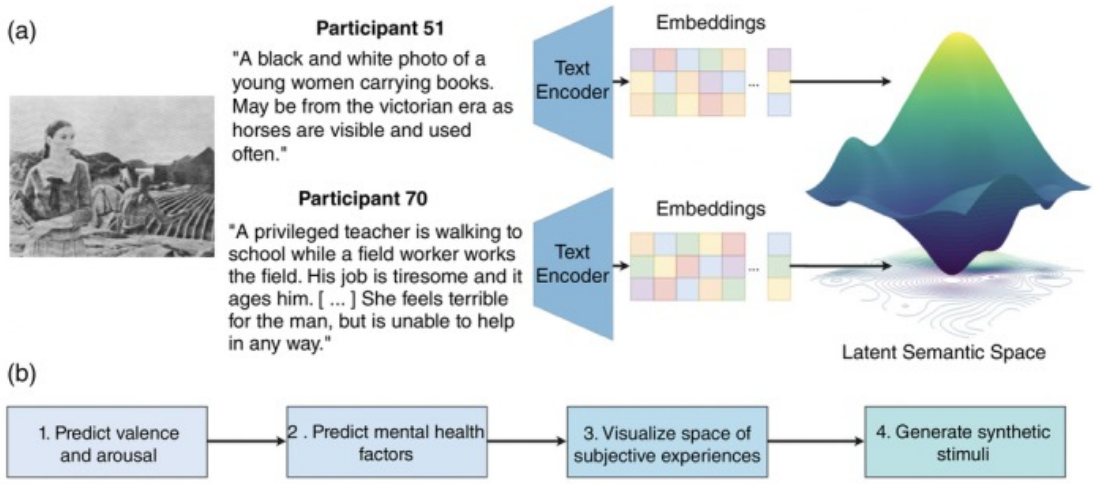


Figure 1: *Mapping subjective experience into semantic space.* In this illustrative visualization (adapted from Manuel et al., 2025), each point represents a narrative response by a participant to an ambiguous stimulus, embedded in a high-dimensional semantic space and projected onto two principal dimensions. Clusters of points indicate semantically related interpretations. For example, responses focusing on *academic themes* (blue cluster, top-right) are located in a different region than those focusing on *physical labor and fatigue* (orange cluster, bottom-left), reflecting distinct attractors of meaning evoked by the same image. Such maps make abstract semantic differences tangible, allowing participants to *see* the landscape of their own mind[41][1].

By examining this map, participants can identify where certain ideas or themes lie. They might notice, for example, that their descriptions involving *personal memories* cluster far from those involving *analytical reasoning*, or that emotionally positive narratives occupy a different region than negative ones. This externalization of one's semantic field is itself a powerful metacognitive tool, as it translates subjective distinctions into a spatial layout[1][42].

3. Interactive Semantic Probing: Next, we introduce **prompt perturbation strategies** analogous to those used in probing LLMs. In LLM research, one might test a model's

understanding by adding a small twist to a prompt and seeing how the completion changes, or by injecting a certain keyword to observe its influence on the output. We will do the human equivalent by guiding participants through a series of modified prompts and instructions, and observing how their responses (and their embedding positions) shift.

- *Contextual Frame Perturbation:* Participants are asked to respond to a prompt from different assigned perspectives or contexts. For example, the base prompt might be "Describe what you see in this image." We then ask them to describe the image "as if you are a worried parent," or "as if you are a detective," or "from a neutral scientific perspective." These frames act like constraints that highlight certain aspects of the image (much as an LLM prompt conditioning on a persona alters its continuation). We predict that the content of the descriptions will change in systematic ways – perhaps the "worried parent" frame yields more emotionally toned and safety-related descriptions, shifting the embedding closer to points associated with anxiety or care, whereas the "scientist" frame yields descriptions focusing on objective details, shifting embeddings towards more neutral, fact-oriented regions. By measuring the displacement in semantic space due to these frame changes, we effectively *map the effect of identity/context on meaning*. This directly tests the idea that identity and social context correspond to directions or axes in semantic space along which thoughts move.
- *Keyword or Cue Injection:* We will also emulate adversarial or bias probes used in LLM studies. For example, in an LLM one can insert a word like "although" or "unexpectedly" to see if it flips the sentiment or outcome of a generated story. For human participants, we can provide a subtle linguistic cue or priming word before they produce a response and see how it biases their interpretation. Concretely, we might show an ambiguous scene and

preface the question with a sentence: “Some people think this scene is humorous.” versus “Some people find this scene tragic.” and then ask the participant to describe it. Even though the image is the same, the hint can bias their mental attractor – a kind of *prompt injection* into their cognitive process. We expect, based on social psychology research on framing effects and priming, that these cues will measurably influence the semantic content of the responses (e.g., more positive or negative language, different thematic focus). Our semantic space analysis would show these as shifts toward the region representing humor or tragedy, respectively. The participants may or may not be consciously aware of the influence. We will debrief them to gather subjective reports on whether they noticed a different mindset.

- *Attentional Focus Guidance:* Another interpretability-inspired approach is analogous to visualizing attention weights in transformers. For humans, we can’t read off neural attention weights easily, but we *can* instruct participants to pay attention to certain aspects of their experience and then report how their description changes. For instance, after a first description of an image, we could say: “Now, focus on the background elements you might have overlooked,” or “Consider the emotions of the people in the scene,” and then have them describe it again. By doing this, we are effectively manipulating the *spotlight of attention*. We predict that deliberately shifting attention will move the semantic content of their report to different attractors (perhaps noticing background elements introduces new story elements – say, shifting from an education theme to an agriculture theme in the earlier example). This exercise not only yields data on how attention alters semantic encoding (attention as a vector that moves us in semantic space), but it also trains participants in introspective control, similar to how one

might train an LLM to attend to a particular part of its context by providing an explicit pointer.

- *Iterative Rephrasing (Chain-of-Thought for Humans):* To mirror chain-of-thought prompting in LLMs (where the model is guided to produce intermediate reasoning steps), we will ask participants to *think out loud* or write down intermediate steps for complex questions. For example, for a moral dilemma question, one group might directly answer, while another is instructed: "List three considerations or angles to this problem, then give your decision." The hypothesis is that forcing a more structured, piecewise articulation (a human analog of an LLM reasoning chain) will lead to more integrative and less biased outcomes. We can examine the semantic content of the intermediate "thought" paragraphs versus the final answer. Does the process of explicating multiple angles lead to a final answer embedding that is more moderate or balanced (perhaps lying semantically between extreme attractors)? If so, this would echo how chain-of-thought in LLMs can reduce reasoning errors – by covering more of the relevant semantic context before collapsing to a conclusion.

4. Data Collection and Measures: Throughout these tasks, we collect (a) the textual responses (for embedding and content analysis), (b) reaction times and any gaze data (if using eye-tracking to see what parts of an image they looked at, relating to attention), and (c) subjective self-reports. After each condition, participants will fill out a brief prompt asking how they felt or what they noticed ("Did the provided perspective influence your description? In what way?", "How difficult was it to adopt the requested mindset?", etc.). The self-reports will help qualitatively interpret the semantic shifts: for example, a participant might write *"Speaking as a 'worried*

parent' made me recall my own child, which shifted how I described the scene" – linking an introspective cause to the observed semantic effect.

5. Analysis: We will analyze the embedding trajectories of each participant's responses across the different perturbations. Techniques from computational linguistics will be used to identify which semantic dimensions (topics, sentiment, formality, etc.) changed significantly under each manipulation. For instance, topic modeling or semantic dimension projection (like evaluating coordinates along predefined axes such as emotional valence) could be applied. We will statistically assess how robustly the perturbations moved responses in the intended direction (e.g., did the "humorous" prime consistently increase the use of positive or comedic language features compared to "tragic" prime, in the majority of participants?). We also plan to compare individual difference factors: perhaps some participants (say, those high in trait openness or cognitive flexibility) show larger semantic shifts when asked to take a new perspective, whereas others remain closer to their original description regardless of prompt. Such findings would be valuable in understanding how *flexible versus rigid* one's semantic attractors are – conceptually akin to the width of attractor basins in a neural network (a very deep basin might be harder to perturb the system out of). In psychological terms, this could correlate with measures like openness to experience or cognitive rigidity.

We will further correlate the quantitative semantic shifts with subjective reports of insight or difficulty. If a participant reports *"I never thought of looking at it that way"* after a perspective shift, and we see a large movement in their semantic representation, it suggests that the experiment facilitated a genuine expansion of their cognitive view for that scenario.

6. Extension to Neuroimaging (optional): While not required for the core of this proposal, an intriguing extension is to conduct a parallel experiment with neuroimaging (fMRI or EEG) to see

if we can identify neural signatures of these semantic space movements. Prior work has shown that combining text embeddings with brain data can reveal how the brain's semantic representations are organized[36][35]. For example, one could present ambiguous stimuli in an fMRI session and have participants adopt different mindsets internally (without necessarily speaking). By using encoding models that project brain activity to the same semantic embedding space, we could test whether the brain's activity patterns move in alignment with the semantic shifts observed in overt behavior. A successful alignment – say, the neural pattern for the "worried parent" perspective lies closer (in the brain representational space) to neural data from known anxiety-related thoughts – would further strengthen the bridge between the LLM-inspired semantic mapping and actual neural processes.

Through this multi-faceted experimental approach, we anticipate not only demonstrating the analogues between prompt-driven changes in LLM outputs and context-driven changes in human interpretations, but also providing participants with a direct experience of their own cognitive flexibility. In effect, participants become researchers of their own minds, using AI-based tools as a mirror. This method could be refined into a form of metacognitive training – for instance, a future app or educational program might let users "play" with their semantic map to see how thinking in different ways literally moves them closer to or farther from certain concepts (e.g., "If I frame my setback as a challenge rather than a failure, I see my narrative point moves into a more positive semantic region"). The experimental protocol here is a first step toward that vision, intended to generate rigorous data on the subjective and objective realities of semantic field navigation.

Implications for Self-Actualization, Metacognition, and Cognitive Design

Embracing the model of the mind as motion through a semantic field carries profound implications. It offers a new language for individuals to understand themselves and for societies to design environments that promote cognitive flourishing. Here we outline several key implications:

- **Empowering Self-Actualization:** If individuals come to see their thoughts and identities as points in a mutable semantic space, they may realize that change and growth are possible by exploring new regions of that space. In humanistic psychology, *self-actualization* refers to fulfilling one's potential and integrating the self into a coherent whole. The semantic paradigm provides practical tools for this journey. For example, someone who has long identified with a narrative "I am not a creative person" may discover, through guided semantic exploration, that this narrative is one attractor in their mind reinforced by specific memories and words. By consciously introducing new contexts – perhaps recalling instances of past creativity or adopting the perspective of a supportive mentor (even via an AI acting as that mentor) – the person can *shift their semantic coordinates* and destabilize the old attractor. This is analogous to cognitive reframing in therapy, but our framework makes it more concrete: reframing is literally moving to a different part of the semantic field where a different set of associations and conclusions hold sway. Over time, with practice, the new perspective can become an attractor in its own right (a stable schema of "I can be creative"). Additionally, understanding the mechanics of semantic attractors can help individuals avoid self-limiting thought patterns. Many *cognitive distortions* (a concept in CBT for erroneous, self-defeating thought patterns) can be seen as getting stuck in a narrow attractor – for

instance, *catastrophizing* zooms in on worst-case interpretations. By visualizing alternative continuations or meanings (much as an LLM generates multiple endings to a sentence), one can counteract the tunnel vision of a distortion. In effect, treating one's negative thought as just one *probabilistic sample* from a distribution of possible thoughts about a situation can lessen its perceived absolute truth. This aligns with mindfulness practices where one observes thoughts as passing mental events rather than facts. Our paradigm gives a computational twist to that idea: the mind is an engine generating many possibilities, and what we experience is the one that "collapsed" into attention – but we can cultivate the ability to consider other samples. This mindset is inherently empowering, as it reinforces that no single narrative is fixed and that we have the freedom to explore and edit our own stories.

- **Advancing Metacognition and Education:** Metacognition – thinking about one's thinking – is crucial for effective learning and decision-making. By introducing the semantic field model in educational settings, we could enhance students' metacognitive skills. For instance, a student solving a problem might learn to recognize when they are stuck in a suboptimal attractor (e.g., approaching a physics problem with a rote formula that doesn't fit) and deliberately inject some randomness or a change of perspective (e.g., thinking analogically by relating the problem to a different domain) to find a new approach. Educators could use visualization tools to show how an idea can be expressed in different ways or how an argument covers only a small region of the conceptual space, prompting critical thinking about what might be missing. In reading comprehension, students could map out the semantic field of a text – identifying clusters of themes or sentiments – to better understand its meaning. This could be particularly helpful in fields

like literature or history, where understanding multiple perspectives is key. Moreover, teaching students that knowledge is not just a collection of facts but a landscape of interconnected concepts could encourage integrative learning. They might see their curriculum as building a mental *map*, and gaps in understanding as simply unexplored territories rather than personal failings. This cartographic metaphor for knowledge aligns well with techniques like concept mapping used in education, but supercharged by AI's ability to handle complexity: imagine an interactive "semantic atlas" that a student can query, powered by an LLM, to show how their current understanding relates to expert knowledge, highlighting paths for growth. On a higher level, framing group discourse in semantic space terms could improve communication. In a debate, participants might realize that they are using the same words but meaning different things (occupying different semantic neighborhoods). By visualizing or explicitly clarifying those differences, they can avoid talking past each other. This is essentially providing a tool for *semantic alignment* in collaborative thinking. In summary, metacognitive training informed by LLM analogies can make abstract skills like reflection, perspective-taking, and conceptual flexibility more tangible and practicable.

- **Cognitive Design in Technology and Society:** The integration of AI's semantic modeling with human cognitive design opens new frontiers in how we build tools and structures for thinking. One application is in mental health technology: digital therapeutics could incorporate semantic mapping to help users track and alter their thought patterns. For example, a journaling app could employ an LLM to analyze the user's daily writings, identify dominant semantic attractors (topics, emotional tones), and gently suggest alternative viewpoints or prompts that lead the user into a different,

perhaps healthier semantic region (e.g., nudging someone ruminating on negatives to also recall positives). Because the suggestions would be personalized via the user's own language (much like an auto-complete offering a different ending to your sentence), this might overcome resistance that direct advice often faces; it's more like discovering a new completion to one's train of thought. On a societal level, understanding minds as navigating semantic spaces could inform how we design information ecosystems. Social media algorithms, for instance, currently exploit attractors in the space of attention – often reinforcing whatever keeps users engaged (which can be outrage or sensationalism). Recognizing this, we could aim to redesign algorithms to facilitate *semantic diversity* and exploratory moves. Rather than trapping users in one attractor bubble (the oft-cited "echo chamber"), platforms could introduce well-calibrated entropy – content that is just different enough to broaden perspectives without being dismissed outright. This idea has echoes in existing calls for "serendipity" in recommendations, but our framework provides a more systematic way to think about it: we want to encourage users' semantic trajectories to occasionally jump gaps and visit other regions of the cultural semantic space, fostering understanding and creativity. Another domain is organizational design and innovation. Companies and research teams thrive when they can balance exploitation of known good ideas and exploration of new ones. By modeling team knowledge as a collective semantic space (perhaps aggregated from their communications, documents, etc.), managers could identify when thinking is too siloed or convergent and then intervene – maybe by bringing in someone with a different semantic perspective, or using an AI facilitation tool that introduces alternative scenarios in meetings (like an "AI devil's advocate" that presents viewpoints underrepresented by the group's current

trajectory). Over time, an organization could cultivate a culture where semantic exploration is valued and guided in strategic ways, leading to more robust and innovative outcomes.

- **Reframing Consciousness and Ethics:** At a philosophical level, our paradigm invites people to reconceptualize what the "self" is. Instead of a static essence or a homunculus in the head, the self becomes the dynamic center of gravity of one's semantic field – a continuity of experience that is nonetheless constantly in flux as it moves through different attractors (roles, emotional states, contexts). This resonates with certain contemplative traditions (Buddhism, for instance) that view the self as a process rather than a thing. By providing a scientific model that parallels that idea, we may offer skeptics a bridge to understand and perhaps adopt such a view. Realizing the fluidity of the self could engender greater empathy: if we appreciate that our perspective at any moment is just one location in a vast landscape, we might find it easier to acknowledge the validity of others' perspectives (they simply occupy a different location) and even to step into their shoes (navigating to their region). This might have ethical implications for reducing conflict and polarization, essentially by promoting a kind of semantic humility – an awareness that no single point of view is all-encompassing and that we can move toward one another through dialogue and open-minded exploration. In the realm of AI ethics, understanding these parallels also urges caution and reflection. If LLMs mimic human-like semantic cognition, what does that imply about their potential to have mind-like properties or to impact us psychologically? For example, people might form attachments or derive authority from LLM outputs because it *feels* like talking to another mind. Our framework can help clarify what is real similarity (e.g., shared structure in

semantic processing) and what is not (LLMs do not have lived experience or emotions, though they can simulate them). Such clarity is vital in discussions about AI consciousness or rights – we can acknowledge the fascinating convergence of pattern processing while maintaining the distinctions in embodiment and agency. At the same time, if we begin to use LLM-based tools as cognitive mirrors or coaches, privacy and autonomy issues arise: these tools might know a great deal about our semantic patterns. Safeguards will be needed to ensure this intimate insight is used for the user's benefit and under their control, not for manipulation. In essence, by blending our cognitive processes with AI, we must consciously *design* how that blending occurs, keeping human values at the center.

In conclusion, the implications of viewing human cognition through the lens of LLM-like semantic dynamics are expansive. They range from the deeply personal – how one might attain self-knowledge and growth – to the structural – how we might redesign societal systems of learning, communication, and AI integration. What unites these diverse threads is the idea of *cognitive alignment*: aligning individuals with healthier, broader semantic horizons, aligning groups with shared understanding despite different starting points, and aligning AI tools with the genuine needs of human cognitive development. We are essentially proposing a co-evolutionary path where insights from artificial models inform human self-improvement, and the richness of human cognitive theory guides ethical AI development.

Conclusion

We have proposed that the mechanics of transformer-based language models offer more than just an engineering advance – they provide a powerful metaphor and scientific framework for re-imagining human cognition and consciousness. By identifying one-to-one correspondences

between LLM operations and cognitive phenomena, we built the case that humans can fruitfully understand their own minds as navigating a semantic space shaped by attractors of meaning, much like an LLM traverses its latent space during text generation. This paradigm synthesizes knowledge from AI, psychology, neuroscience, and sociology, pointing toward a unified conceptual vocabulary for minds both artificial and biological.

The cross-domain mappings we presented – from semantic attractor fields to semantic memory networks, from probabilistic token collapse to perceptual decision-making, from entropy modulation to cognitive flexibility, and from constraint-driven emergence to goal-oriented thought – underscore a deep structural similarity in the dynamics of thought and language. LLMs, in a very real sense, operationalize a model of meaning-making: they demonstrate how syntax and semantics can arise from statistical associations, how context can direct attention to relevant information, and how new capabilities can spontaneously emerge from complexity. Human cognitive science, on the other hand, provides rich descriptions and theories (predictive processing, global workspaces, social construction of meaning) that can now be seen reflected in these artificial models[36][35]. This bidirectional illumination – using AI to understand cognition and cognition to interpret AI – is an exciting development in the philosophy of mind and the science of intelligence.

We put forward not just a theoretical bridge, but also a practical method to explore it: an experimental paradigm where individuals engage in *introspective semantic experiments* with the aid of AI-based tools. By visualizing one's thoughts as points in a semantic map and observing how they move with changes in perspective or context, people can gain a novel form of self-insight. The initial protocol we described can be expanded in future work – for example, into longitudinal studies where participants practice semantic perspective-shifting as a skill and

researchers measure impacts on creativity, empathy, or mental well-being. The potential for therapeutic and educational applications is considerable. If empirical results continue to support our framework – showing that subjective experiences and transformations can be rigorously tracked in a semantic space – it could catalyze new fields like *computational phenomenology* or *AI-augmented introspection*, in which subjective reports and objective data are integrated[39][1].

There are, of course, important caveats and open questions. Our paradigm, while evocative, simplifies many aspects of biology and consciousness. Human emotions, for instance, are only loosely captured by semantic content – they also involve embodiment, physiology, and unconscious drives. Future refinements might integrate those dimensions (for example, adding physiological measurements to see how semantic moves correlate with emotional arousal). Moreover, not all features of LLMs have clear cognitive analogues: large models can sometimes produce spurious text that humans would never generate, and humans have motives and desires that an LLM lacks. We must be careful not to over-equate the two. Rather, the value of the comparison is in providing *models* and *metaphors* that can be empirically examined. In the spirit of cognitive science, LLMs serve as working models of certain aspects of thought – not complete replicas of the mind, but fertile ground for hypotheses. By studying their limitations (e.g., areas where LLMs fail at reasoning or common sense that humans find trivial), we also sharpen our understanding of what additional mechanisms our brains employ (perhaps grounding in physical experience or emotional coloring) that pure language prediction misses[43][44]. This can drive the next generation of models to include those elements, bringing AI closer to human-like understanding, while simultaneously giving cognitive science a more comprehensive theoretical toolkit.

In closing, we envision a future in which the study of natural and artificial intelligence becomes increasingly interwoven. The paradigm shift we advocate – seeing the self as a navigator of semantic space – could transform practices from psychotherapy to education to human-computer interaction. Just as the cognitive revolution of the mid-20th century empowered psychology with concepts from computation (like information processing and memory buffers), the current revolution in deep learning offers new constructs (like attention heads and embedding spaces) to enrich our theories of mind. Embracing this opportunity, however, requires an interdisciplinary openness and a willingness to translate insights across fields. We hope this paper serves as a step in that direction, building bridges between the silicon and the synapse, between data-driven models and lived experience.

Ultimately, the measure of this framework's worth will be in what it enables – new experiments, new interventions, and perhaps most importantly, new ways for individuals to relate to their own minds. By scientifically validating that we can introspect in terms of attractors and trajectories, we hand people a map and compass for the terrain of their consciousness. The paradigm shift is then not only in understanding, but in being: empowering each person to become an explorer and author of their semantic world, in a journey toward greater self-actualization, empathy, and wisdom.

References: *(Selection)*

- Clark, A. (2013). Whatever next? *Predictive brains, situated agents, and the future of cognitive science*. *Behavioral and Brain Sciences, 36*(3), 181–204.[10]
- Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience, 11*(2), 127–138.[15][23]

- Ramsauer, H., et al. (2021). Hopfield Networks is All You Need. *Proceedings of ICLR 2021*. (Demonstrates transformer attention as Hopfield memory retrieval)[11][12]
- McClelland, J. L., & Rogers, T. T. (2003). The parallel distributed processing approach to semantic cognition. *Nature Reviews Neuroscience, 4*(4), 310–322. (Attractor networks in semantic memory)[2][5]
- Shaki, J., et al. (2023). Cognitive effects in large language models. *Proc. European Conf. on AI*. (Reports GPT-3 exhibiting human-like priming and bias effects)[45]
- Schrimpf, M., et al. (2021). The neural architecture of language: Integrative modeling converges on predictive processing. *PNAS, 118*(45), e2105646118. (Aligns transformer models with brain activity)[18]
- Manuel, S., et al. (2025). Towards a latent space cartography of subjective experience in mental health. *Psychiatry and Clinical Neurosciences, 79*(5), 248–260. (Maps participant narratives to latent semantic space)[39][1]
- Berger, P. & Luckmann, T. (1966). *The Social Construction of Reality*. (Classic sociology text on how meaning is co-constructed in society)[9]
- Baars, B. (1988). *A Cognitive Theory of Consciousness*. (Introduces Global Workspace Theory)[32]
- Miller, E. & Cohen, J. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience, 24*, 167–202. (Top-down control biasing processing)[20][21]
- Wei, J., et al. (2022). Emergent abilities of large language models. *Transactions on Machine Learning Research*. (Discusses unpredictable capabilities in scaled LLMs)[19]

- Caucheteux, C. & King, J.-R. (2022). Brains and algorithms partially converge in natural language processing. *Communications Biology, 5*, 134. (Brain-LM representational similarity)[46]
- Marjieh, R., et al. (2023). Large language models predict human sensory judgments across modalities. *arXiv:2303.06733*. (GPT models capturing perceptual similarity)[47]
- Suresh, S., et al. (2023). A cognitive science investigation of conceptual representations in LLMs. *arXiv:2308.07820*. (Compares stability of concepts in humans vs LLMs)[48]
- Vygotsky, L. (1987). *Thinking and Speech*. (Original work 1934; emphasizes social origin of thought)

(Additional references included in text citations above.)

---

[1] [34] [35] [36] [39] [40] [41] [42] Towards a latent space cartography of subjective experience in mental health - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC12047060/

[2] [5] [13] [29] [30] [31] An attractor model of lexical conceptual processing: simulating semantic priming - ScienceDirect
https://www.sciencedirect.com/science/article/abs/pii/S0364021399000051

[3] [4] [17] [18] [27] [28] [37] [38] [43] [44] [46] [47] [48] Large Language Models and Cognitive Science: A Comprehensive Review of Similarities, Differences, and Challenges
https://arxiv.org/html/2409.02387v1

[6] [8] [14] Itinerancy between attractor states in neural systems - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC5056802/

[7] [PDF] Optimal Foraging in Semantic Memory - Cognitive Science

https://www.cogsci.msu.edu/DSS/2019-2020/Jones/HillsetalPsychRev2012.pdf

[9] Mind, Self, and Society by George Herbert Mead | Research Starters

https://www.ebsco.com/research-starters/literature-and-writing/mind-self-and-society-george-herbert-mead

[10] Whatever next? Predictive brains, situated agents, and the future of ...

https://pubmed.ncbi.nlm.nih.gov/23663408/

[11] [12] [2008.02217] Hopfield Networks is All You Need

https://arxiv.org/abs/2008.02217

[15] [16] [23] [24] [25] [26] The free-energy principle: a unified brain theory?

https://www.uab.edu/medicine/cinl/images/KFriston_FreeEnergy_BrainTheory.pdf

[19] [PDF] Emergent abilities of large language models - arXiv

https://arxiv.org/pdf/2206.07682

[20] An Integrative Theory of Prefrontal Cortex Function - ResearchGate

https://www.researchgate.net/publication/12050870_An_Integrative_Theory_of_Prefrontal_Cortex_Function

[21] An integrative theory of prefrontal cortex function. - Semantic Scholar

https://www.semanticscholar.org/paper/An-integrative-theory-of-prefrontal-cortex-Miller-Cohen/29c7b2ff6c728034fc444df24da39ffebf35ea9e

[22] [PDF] An integrative theory of prefrontal cortex function.

https://ekmillerlab.mit.edu/wp-content/uploads/2025/06/Miller-Cohen-2001.pdf

[32] [33] Global workspace theory - Wikipedia

https://en.wikipedia.org/wiki/Global_workspace_theory

[45] Meaningless is better: Hashing bias-inducing words in LLM prompts ...

https://www.sciencedirect.com/science/article/abs/pii/S0306457325004340